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| **Fiscal Year:** | FY 2009 | **Task Last Updated:** | FY 03/12/2009 |
| **PI Name:** | Maurer, Richard H. Ph.D. | | |
| **Project Title:** | Combined Ion and Neutron Spectrometer for Space Applications | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | NSBRI | | |
| **Program/Discipline--Element/Subdiscipline:** | NSBRI--Smart Medical Systems and Technology Team | | |
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| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 20723-6099 | **Congressional District:** | 7 |
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| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2003 Biomedical Research & Countermeasures 03-OBPR-04 |
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| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NSBRI |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 11/30/2008 per NSBRI (8/2008) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Goldsten, John ( The Johns Hopkins University Applied Physics Laboratory )<br>Roth, David ( The Johns Hopkins University Applied Physics Laboratory )<br>Zeitlin, Cary ( Lawrence Berkeley National Laboratory )<br>Haggerty, Dennis ( The Johns Hopkins University Applied Physics Laboratory ) | | |
| **Grant/Contract No.:** | NCC 9-58-TD00404 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

**Task Description:**

(1) Original Aims
We proposed to design and fabricate a Combined Ion and Neutron Spectrometer (CINS) for Space Applications. This instrument improves upon existing charged particle species and energy spectrometers (e.g., MARIE on Mars Odyssey) and incorporates the capability of the previously developed NSBRI Neutron Energy Spectrometer to yield a single and complete ionizing radiation environment monitor for application in space habitats and transport vehicles. The originally proposed objectives were refined with respect to specific metrics in February/March 2005 to a) eliminate the gain saturation of MARIE for heavy ions with LET > 35 keV/micron; b) increase the dynamic range vs. the MARIE instrument by a factor of 100-200 to a minimum of 10,000:1 to include protons with energies above 100 MeV; c) increase the maximum instrument event detection rate by at least a factor of 10 to a minimum of 100 Hz.

The end product of the effort was to be a Technology Readiness Level 6 instrument.

(2) Key Developments

The conceptual design of the CINS instrument was facilitated through the use of Monte Carlo simulations that modeled the basic detector stack, established the energy resolution, and quantified the issues that would affect the instrument's efficiency using GEANT4. Our CINS design consists of a stack of 7 cylindrical detectors with a radius of 20 mm. There are 3 different types of detectors in the CINS stack. One is a standard Silicon (Si) solid state detector. The remaining two detector types are scintillators. The first detector in the stack is a 1 mm thick BC430 detector followed by a 5 mm thick Si detector, then a second 1 mm BC430 detector, followed by three 5 mm Si detectors. The final detector is a 38 mm thick scintillator detector of BGO.

For the CINS system to accurately resolve solar energetic particle (SEP) events the system must cleanly separate the protons from the electrons. We simulated spectra of electrons to 20 MeV and protons to 1 GeV. While a small percentage of the electrons and protons do overlap in this simulation, the vast majority of the protons can be separated from the electrons. Protons up to ~150 MeV are stopped in the stack. Given reasonable energy resolution of the BGO scintillator the simulation showed that protons up to ~300 MeV can be uniquely identified.

We re-drifted eight Si(Li) detectors for charged particles and neutrons from Lawrence Berkeley Laboratory's existing stock. The second step was the creation of a guard ring accomplished by routing out the detector in a circular pattern at a fixed radius. These eight detectors have been evaluated in experiments at HIMAC and NSRL during 2006 and 2007. For the charged particle telescope we procured a 38mm X 38mm BGO scintillator detector unit and obtained 1mm thick scintillators. We designed and fabricated detector boards for NSRL evaluations and designed charged particle telescope electronics for control, power and data acquisition.

Ground based accelerator experiments were executed at NSRL in March 2006 by colliding a 1 GeV/amu iron beam with a 5 inch thick aluminum target simulating a large space vehcle. The progeny of these collisions, both charged particle fragments and neutrons, were monitored by the refurbished 5mm thick silicon detectors. During May 2007 five of the seven detector elements of the charged particle telescope were operating and a large volume of charged particle fragment data was obtained. For the 2007 experiments iron and silicon beams and targets of 10-20 g/cm^2 were used. All seven elements of the charged particle telescope were active for the first time in Fall 2007. Additional experimental runs were made in Spring, Summer and Fall 2008. For the Neutron Energy Spectrometer JHU/APL designed, procured and fabricated a 12cm X 12cm Eljen plastic boron-loaded scintillator detector for medium energy 1-14 MeV neutrons that have the maximum dose equivalent weighting factors of 10-20. This fast neutron detector system was calibrated at Columbia University RARAF in November 2006 and used at NSRL for the first time in May 2007. This scintillator is operated in addition to the existing 5mm thick silicon detectors used for high energy neutron spectra above 10 MeV. The silicon detectors must be used in conjunction with standard paddle scintillators to veto the charged particle fragments.

A custom data acquisition system was also designed and fabricated for the charged particle telescope during 2007. The output of the acquisition system has a large usable dynamic range and is able to handle the relatively high event rates that occur in an accelerator beam. This system detects events, proceeds to capture waveform data from the preamplifiers and saves the data to a hard drive. The system is built around several digital to analog converters and a field programmable gate array and is essentially a fast 16 channel digital oscilloscope. It recorded data for the NSRL experiments beginning in Fall 2007.

We have succeeded in achieving all our original aims with respect to the instrument hardware. We also were able to produce data on charged particle fragment composition and energy spectra and neutron energy spectra from several collision experiments. A sample of the data has been published in the journal Nuclear Instruments and Methods B. A second paper in the same journal describes the custom data acquisition system.

(3) Impact of Development

As built CINS consists of a) high and medium energy neutron detectors; b) a charged particle detector telescope based on a standard silicon stack concept and c) a custom data acquisition system. CINS meets all the aims defined in (1). Its subsystems have all been successfully tested and calibrated with accelerator beams. The instrument has a technology readiness level of 6. A space flight quality instrument package can be easily produced by reducing volume, mass and power consumption.

**Rationale for HRP Directed Research:**

**Research Impact/Earth Benefits:**

Neutron detection for homeland security is of interest in the areas of 1) the location of radiological weapons used by terrorists, 2) scanning and inspecting imported cargo in ship holds or on the dock side and 3) smart portals for the scanning of personnel or small packages at selected points of entry.

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| **Task Progress:** | Task progress over the last funded year (~FY 2008)<br>Charged Particle Telescope<br><br>For the charged particle telescope we procured a 38mm X 38mm BGO scintillator detector unit and obtained 1mm thick scintillators. We designed and fabricated detector boards for NSRL evaluations and designed charged particle telescope electronics for control, power and data acquisition. All telescope hardware was completed by Fall 2007 when the seven element telescope was completely evlauted for the first time at NSRL. Additional experiments and evaluations with and without targets were performed during the Spring 08, Summer 08 and Fall 08 NSRL runs. Several different material targets were used in the collision experiments in the range of 2-20 g/cm^2.<br><br>Data Acquisition System<br><br>The CINS data acquisition system was also completed and used at NSRL for the first time in Fall 2007. This system includes the use of ADCs (Analog to Digital Converters) and an FPGA (Field Programmable Gate Array) which is essentially the heart of a fast 16 channel digital oscilloscope. The system detects events (comparing incoming signals to user-supplied trigger definitions), proceeds to capture waveform data from the preamplifiers and save the data to a hard drive. It can be used with both the charged particle and neutron elements of CINS.<br><br>Neutron Energy Spectrometer<br><br>For the Neutron Energy Spectrometer JHU/APL designed, procured and fabricated a 12cm X 12cm Eljen plastic boron-loaded scintillator detector for medium energy 1-14 MeV neutrons that have the maximum dose equivalent weighting factors of 10-20. This detector system was calibrated at Columbia University RARAF in November 2006 and used at NSRL for the first time in May 2007. Several attempts in 2007 and 2008 were made to evaluate the scintillator at NSRL but the gamma background in the room proved to be too high to get meaningful neutron data in the 1-14 Mev energy range.<br><br>The silicon detectors with scintillator paddle charged particle discrimination did produce useful high energy neutron data (>10 MeV)in the NSRL collision runs with thick targets.<br><br>Publications<br><br>Two papers, one on the CINS data acquisition system and one on the development and evaluation of CINS (including much charged particle and neutron data from NSRL), were accepted for publication by the journal Nuclear Instruments and Methods B.<br><br>Summary<br><br>We basically have successfully completed what we set out to do---the design, development, fabrication and evaluation of CINS that produced a working laboratory instrument and publishable charged particle and neutron data from NSRL experiments. |
| **Bibliography Type:** | Description: (Last Updated: 04/27/2011) |
| **Articles in Peer-reviewed Journals** | Grey M, Goldsten J, Maurer R, Roth D, Zeitlin C. "Data acquisition for the Combined Ion and Neutron Spectrometer (CINS)." Nuclear Instruments and Methods in Physics Research Section B: Beam Interactions with Materials and Atoms 2009 Jan;267(1):139-43. http://dx.doi.org/10.1016/j.nimb.2008.10.093 , Jan-2009 |
| **Articles in Peer-reviewed Journals** | Zeitlin C, Maurer R, Roth D, Goldsten J, Grey M. "Development and evaluation of the Combined Ion and Neutron Spectrometer (CINS)." Nuclear Instruments and Methods in Physics Research Section B: Beam Interactions with Materials and Atoms, 2009 Jan;267(1):125-38. http://dx.doi.org/10.1016/j.nimb.2008.10.006 , Jan-2009 |